

PROJECT IDEAS & TIERS

How to Choose Your Project Tier

Tier 1: CORE PROJECT (100 points possible)

- **For:** Everyone, especially beginners
- **Difficulty:** ★ ★ ☆ ☆ ☆
- **Time:** Achievable in 5-6 weeks
- **Resources:** Free tier (Colab/Kaggle)
- **Description:** Simple CV project using pre-trained models

Tier 2: ADVANCED PROJECT (+10 bonus points)

- **For:** Students wanting a challenge
- **Difficulty:** ★ ★ ★ ☆ ☆
- **Time:** Requires consistent work
- **Resources:** May need Heidi Platform or small API budget (\$5-10)
- **Description:** Multi-component system or VLM/Agentic project

Tier 3: EXCEPTIONAL PROJECT (+20 bonus points)

- **For:** Experienced students or very motivated teams
- **Difficulty:** ★ ★ ★ ★ ★
- **Time:** Requires significant effort
- **Resources:** Creative use of free resources + some paid APIs
- **Description:** Production-quality or research-level system

Recommendation: Start with Tier 1. You can always add features to reach Tier 2!



TIER 1 PROJECT IDEAS

Simple, achievable projects perfect for learning CV fundamentals.

1. Product Detection for Inventory

What: Detect and count products on shelves

Model: YOLOv8 (pre-trained, fine-tune on your products)

Data: Take photos with your phone OR use Roboflow dataset

Output: Annotated images with counts

Real-world use: Warehouse inventory management

Difficulty: ★★☆☆☆

Why it's great:

- Easy to collect data (just photograph items)
 - Clear success metric (counting accuracy)
 - Immediate practical value
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2. Face Mask Detection

What: Detect if people are wearing masks correctly

Model: YOLOv8 or Faster R-CNN

Data: Public datasets from Kaggle (thousands of images)

Output: "Mask" / "No Mask" / "Incorrect Mask"

Real-world use: Public health compliance monitoring

Difficulty: ★★☆☆☆

Why it's great:

- Public datasets readily available
 - Socially relevant problem
 - Easy to test on your own photos
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3. Plant Disease Classification

What: Identify diseases in plant leaves

Model: ResNet50 or EfficientNet (ImageNet pre-trained)

Data: PlantVillage dataset (54,000+ images, free)

Output: Disease name and confidence score

Real-world use: Help farmers identify crop diseases

Difficulty: ★★☆☆☆

Why it's great:

- Excellent public dataset
 - Clear classification task
 - Helpful for agriculture
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4. License Plate Detection

What: Detect license plates in images

Model: YOLOv8

Data: Public dataset or collect from parking lots

Output: Bounding boxes around plates

Real-world use: Automated parking systems, traffic monitoring

Difficulty: ★★☆☆☆

Why it's great:

- Interesting real-world application
 - Can collect your own data easily
 - Can extend with OCR later (Tier 2)
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5. Animal Species Classification

What: Classify different animal species

Model: ResNet or EfficientNet

Data: iNaturalist dataset or custom wildlife dataset

Output: Species name and probability

Real-world use: Wildlife monitoring, biodiversity research

Difficulty: ★★☆☆☆

Why it's great:

- Fun and visually interesting
 - Large datasets available
 - Good for learning transfer learning
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6. Waste Classification (Recycling Helper)

What: Classify waste into recyclable categories

Model: MobileNet or EfficientNet

Data: TrashNet dataset or collect your own

Output: "Plastic" / "Paper" / "Metal" / "Glass" / "Trash"

Real-world use: Smart recycling bins

Difficulty: ★★☆☆☆

7. Traffic Sign Recognition

What: Detect and classify traffic signs

Model: YOLOv8 + Classification head

Data: German Traffic Sign Recognition Benchmark (GTSRB)

Output: Sign type and location

Real-world use: Autonomous vehicles

Difficulty: ★★☆☆☆

8. Food Classification

What: Identify food items in photos

Model: EfficientNet or ResNet

Data: Food-101 dataset (101 food categories)

Output: Food name and nutritional info

Real-world use: Calorie tracking apps

Difficulty: ★★☆☆☆



TIER 2 PROJECT IDEAS

More complex systems requiring multiple components or advanced techniques.

1. Visual Question Answering System

What: Answer natural language questions about images

Model: BLIP or LLaVA (via Hugging Face)

Data: VQA v2 dataset or custom images

Example: Image of kitchen → Q: "How many chairs?" → A: "4 chairs"

Real-world use: Accessibility for visually impaired

Difficulty: ★★★☆☆

Why Tier 2:

- Combines vision + language
 - Requires understanding VLMs
 - More complex evaluation
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2. Multi-Object Tracking for People Counting

What: Track people in video and count entries/exits

Model: YOLOv8 + DeepSORT

Data: MOT Challenge dataset or record your own videos

Output: Person IDs, trajectories, entry/exit counts

Real-world use: Retail analytics, crowd management

Difficulty: ★★★☆☆

Why Tier 2:

- Requires tracking algorithm
 - Video processing is more complex
 - Needs good temporal reasoning
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3. Image Captioning System

What: Generate natural language descriptions of images

Model: BLIP or CLIP + GPT

Data: COCO Captions dataset

Output: "A person riding a skateboard in a park"

Real-world use: Alt text generation, content moderation

Difficulty: ★ ★ ★ ☆ ☆

Why Tier 2:

- Language generation component
 - Requires understanding encoder-decoder models
 - Complex evaluation metrics (BLEU, CIDEr)
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4. Safety Equipment Detection (PPE)

What: Detect if workers wear hard hats, vests, gloves

Model: YOLOv8 fine-tuned on PPE dataset

Data: Roboflow PPE dataset

Output: Compliance report with violations highlighted

Real-world use: Construction site safety monitoring

Difficulty: ★ ★ ★ ☆ ☆

Why Tier 2:

- Multiple object classes
 - Requires good training/fine-tuning
 - Real safety implications
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5. Semantic Segmentation for Autonomous Driving

What: Segment road, cars, pedestrians, etc.

Model: DeepLabV3 or SegFormer

Data: Cityscapes dataset (free for academic use)

Output: Per-pixel class labels

Real-world use: Self-driving cars

Difficulty: ★ ★ ★ ★ ☆

Why Tier 2:

- More complex than detection
- Requires understanding segmentation

- Large, detailed datasets

6. Document Layout Analysis

What: Detect and classify document regions (title, text, image, table)

Model: LayoutLM or custom detection model

Data: PubLayNet or DocLayNet dataset

Output: Structured document understanding

Real-world use: Document digitization, OCR preprocessing

Difficulty: ★ ★ ★ ☆ ☆

7. Emotion Recognition from Faces

What: Detect faces and classify emotions

Model: YOLOv8 for detection + EfficientNet for classification

Data: FER2013 or AffectNet dataset

Output: "Happy" / "Sad" / "Angry" / etc. with confidence

Real-world use: Customer sentiment analysis, mental health

Difficulty: ★ ★ ★ ☆ ☆

8. Optical Character Recognition (OCR) System

What: Detect text regions and extract text

Model: EAST text detection + Tesseract OCR

Data: ICDAR dataset or custom documents

Output: Extracted text with bounding boxes

Real-world use: Document processing, receipt scanning

Difficulty: ★ ★ ★ ☆ ☆



TIER 3 PROJECT IDEAS

Production-quality or research-level projects.

1. Agentic Visual Assistant

What: AI agent that sees, reasons, and takes actions

Model: VLM (GPT-4V or LLaVA) + LangChain framework

Example: "Agent, check if this workspace is safe" → Analyzes image → Identifies hazards → Generates safety report

Real-world use: Automated safety inspections, quality control

Difficulty: ★ ★ ★ ★ ★

Why Tier 3:

- Combines vision + reasoning + action
 - Requires understanding agent architectures
 - Complex prompt engineering
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2. Multi-Modal Search Engine

What: Search images using text OR find similar images

Model: CLIP for embeddings + FAISS vector database

Data: Your own image collection (1000+ images)

Example: "Red sports car" → Returns matching images

Real-world use: E-commerce, digital asset management

Difficulty: ★ ★ ★ ★ ☆

Why Tier 3:

- Requires vector database setup
 - Scalability considerations
 - Cross-modal retrieval is complex
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3. Real-Time Video Analysis Pipeline

What: Process live video with multiple CV tasks

Components: Detection + Tracking + Classification + Captioning

Model: YOLOv8 + DeepSORT + BLIP (optimized)

Data: Live webcam feed

Real-world use: Surveillance, sports analytics

Difficulty: ★ ★ ★ ★ ★

Why Tier 3:

- Real-time constraints (30+ FPS)
 - Multiple models running simultaneously
 - Optimization and engineering challenges
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4. Visual Reasoning System

What: Answer complex reasoning questions about images

Model: LLaVA or GPT-4V with chain-of-thought

Data: Visual reasoning datasets (NLVR2, GQA)

Example: "Is there more furniture on the left or right?" → Requires spatial reasoning

Real-world use: Advanced visual analytics

Difficulty: ★ ★ ★ ★ ★

Why Tier 3:

- Requires advanced prompting
 - Complex reasoning evaluation
 - Research-level problem
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5. Custom VLM Fine-Tuning

What: Fine-tune a VLM for specific domain

Model: BLIP-2 or LLaVA (7B-13B parameters)

Data: Domain-specific dataset (medical, industrial, etc.)

Example: Medical image analysis with domain knowledge

Real-world use: Specialized industry applications

Difficulty: ★ ★ ★ ★ ★

Why Tier 3:

- Requires significant compute (A100)
 - Understanding of fine-tuning techniques
 - Custom data collection and labeling
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6. 3D Object Detection (Point Clouds or Depth)

What: Detect objects in 3D space

Model: PointNet++ or VoteNet

Data: KITTI 3D or ScanNet dataset

Output: 3D bounding boxes

Real-world use: Robotics, AR/VR

Difficulty: ★ ★ ★ ★ ★

7. Zero-Shot Object Detection

What: Detect objects never seen during training

Model: CLIP + Grounding DINO

Data: Open-vocabulary dataset

Example: Detect "purple llama" without training on llamas

Real-world use: Flexible object detection systems

Difficulty: ★ ★ ★ ★ ★

8. Multi-Camera Multi-Object Tracking

What: Track objects across multiple camera views

Model: Custom re-identification network

Data: Multiple video streams or MTMC dataset

Output: Unified tracks across all cameras

Real-world use: Retail analytics, security systems

Difficulty: ★ ★ ★ ★ ★

HOW TO CHOOSE?

Ask Yourself:

1. What interests me?

- Choose something you're genuinely curious about!

2. What's my experience level?

- First CV project? → Tier 1
- Done ML before? → Consider Tier 2
- Very experienced? → Challenge yourself with Tier 3

3. How much time do I have?

- Limited time? → Tier 1 (safe choice)
- Can dedicate 10+ hours/week? → Tier 2
- Can dedicate 15+ hours/week? → Tier 3

4. Do I have a team?

- Solo? → Stick to Tier 1
- Team of 2-3? → Tier 2 is very doable
- Strong team? → Tier 3 is achievable

5. What resources do I have?

- Only Colab free? → Tier 1
 - Access to Heidi? → Tier 2 possible
 - Can spend \$10 on APIs? → Tier 3 possible
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IMPORTANT REMINDERS

1. Better to do Tier 1 excellently than Tier 3 poorly
2. You can always add features to upgrade tiers later

3. **Tier 1 done well = 100% grade!**
 4. **Bonus points are BONUS - not required**
 5. **Pick something you can demo successfully**
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STILL CAN'T DECIDE?

Default Recommendation for Most Students:

- **Tier 1: Object Detection Project** (YOLOv8)
- Easy to understand, lots of tutorials
- Can always enhance it later
- Guaranteed to work with free resources

If You Want a Challenge:

- **Tier 2: Visual Question Answering** (BLIP)
 - Hot topic in AI right now
 - Good balance of complexity and feasibility
 - Impressive demo potential
- Double-click (or enter) to edit
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Start coding or [generate](#) with AI.

This is formatted as code